



Jinpeng Lu

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World Model / VLM & VLA Representation Learning
Medical Vision Foundation Models



Research Profile & Core Strengths

My research follows the same three areas as my personal homepage and studies how models build, maintain, and transfer reliable representations: **World Model** for dynamic world-state diagnostics, **VLM & VLA Representation Learning** for how multimodal representations become actions, and **Medical Vision Foundation Models** for efficient 3D perception, multimodal segmentation, and transfer limits.

My work emphasizes diagnostic and systems-oriented research: converting open-ended capability questions into reproducible benchmarks, causal/behavioral evidence chains, and efficient model designs, including WRBench dynamic world-state diagnostics, VeloxSeg, DINOv3 medical transfer evaluation, and H2ASeg PET/CT tumor segmentation.

Experienced with PyTorch / DDP / CUDA / Linux training and evaluation pipelines, data construction, custom model implementation, ablation analysis, visualization, and English research writing.

Education

University of Science and Technology of China

Sep. 2025 – Jun. 2028

M.S., Information and Communication Engineering

Advisor: Zhiwei Xiong

Harbin Institute of Technology, Shenzhen

Sep. 2021 – Jun. 2025

B.S. in Mathematics; GPA: 3.728/4, Rank: 17/172, Top 10% Mathematics / Statistics / Numerical Computing

Representative Works

World Model

Current World Models Lack a Persistent State Core, *First Author*

arXiv 2026

- **Task:** Evaluate whether world models maintain dynamic scene state under viewpoint changes, beyond generating locally plausible videos.
- **Method:** Built WRBench with controlled camera motion and a D1-D6 diagnostic profile spanning visible consistency, re-observation support, and returned-state consistency.
- **Result:** Evaluated 23 models on 9,600 videos and found systematic weaknesses in dynamic state maintenance and re-observation consistency.

Medical Vision Foundation Models

VeloxSeg: JL-Guided Efficient 3D Medical Segmentation, *First Author*

ICLR 2026

- **Task:** Make 3D multimodal medical segmentation, including PET/CT and MRI, cheaper to deploy while still modeling context across imaging modalities.
- **Method:** Designed a lightweight network that combines local structure, cross-modal context, and texture cues from self-supervised models.
- **Result:** With only 1.66M parameters, achieved a 26% Dice score gain and improved GPU / CPU throughput by 11x / 48x.

Does DINOv3 Set a New Medical Vision Standard?, *Co-first Author*

arXiv 2025

- **Task:** Evaluate whether DINOv3 can serve as a general medical vision foundation model across structural imaging, pathology slides, electron microscopy, and PET.
- **Method:** Co-built a 2D/3D transfer benchmark covering classification, segmentation, and registration, then compared performance across modalities, task types, and model scales.
- **Result:** Found strong baselines on CT/X-ray but degradation on whole-slide pathology images, electron microscopy, and PET, with unstable scaling behavior.

H2ASeg: Hierarchical PET/CT Tumor Segmentation, *First Author*

MICCAI 2024

- **Task:** Improve PET/CT tumor segmentation where PET highlights tumor location and CT provides anatomical boundaries, but the two signals are hard to combine reliably.
- **Method:** Designed hierarchical cross-modal interaction to align PET semantic cues with CT structure, plus tumor-aware weighting for complex lesion regions.
- **Result:** Outperformed prior methods on AutoPET-II and Hecktor2022; module ablations produced 12.20% / 10.94% Dice score gains.

Core Methods & Engineering

Research: World Model evaluation, VLM/VLA representation diagnostics, Medical Vision Foundation Models transfer, PET/CT and MRI 3D segmentation

Engineering: Python, PyTorch, CUDA, DDP, Shell/Bash, LaTeX; training/evaluation pipelines, experiment reproduction, ablation studies, custom model implementation

Tools: Linux, Git, Docker, Weights & Biases / TensorBoard, visual, multimodal, and medical-imaging data-processing toolchains

Writing: English research writing, experiment analysis, and ICLR / MICCAI / Medical Image Analysis submission experience

Full Research Outputs

World Model

2026 – Present

- [1] **Current World Models Lack a Persistent State Core**, *First Author*

arXiv 2026

Contribution Introduces WRBench for dynamic world-state evaluation under controlled viewpoint changes with a D1-D6 diagnostic profile.

VLM & VLA Representation Learning

2026 – Present

- [1] **VLA-Trace: Diagnosing Vision-Language-Action Models through Representation and Behavior Tracing**, *Co-author*

arXiv 2026

Contribution Connects representation tracing with rollout behavior to diagnose routing, grounding, and semantic-following failures in VLA models.

- [2] **Pelican-Unify 1.0: A Unified Embodied Intelligence Model**, *Co-author*

arXiv 2026

Contribution Contributes to unified embodied model training and evaluation across understanding, reasoning, future imagination, and action generation.

Medical Vision Foundation Models

2023 – 2026

- [1] **Johnson-Lindenstrauss Lemma Guided Network for Efficient 3D Medical Segmentation**, *First Author*

ICLR 2026

Contribution Proposes VeloxSeg for efficient multimodal 3D medical segmentation under a lightweight parameter budget.

- [2] **Does DINOv3 Set a New Medical Vision Standard?**, *Co-first Author*

arXiv 2025

Contribution Evaluates DINOv3 transfer across medical modalities and identifies where foundation-model scaling remains unreliable.

- [3] **H2ASeg: Hierarchical Adaptive Interaction and Weighting Network for Tumor Segmentation in PET/CT Images**, *First Author*

MICCAI 2024

Contribution Improves PET/CT tumor segmentation through hierarchical multimodal interaction and tumor-aware weighting.

- [4] **AttriMIL: Revisiting Attention-based Multiple Instance Learning for WSI Classification**, *Co-author*

Medical Image Analysis 2025

Contribution Revisits attention-based multiple-instance learning for interpretable whole-slide image classification.

- [5] **IPGPhormer: Interpretable Pathology Graph-Transformer for Survival Analysis**, *Co-author*

arXiv 2025

Contribution Studies interpretable pathology graph modeling for survival analysis and prognosis prediction.

- [6] **A Localization-to-Segmentation Framework for Automatic Tumor Segmentation in Whole-Body PET/CT Images**, *Co-author*

arXiv 2023

Contribution Develops a localization-to-segmentation cascade for automatic whole-body PET/CT tumor segmentation.

Additional Project

Generative Sensor Signal Enhancement | *Lead, National Innovation Program*

Jun. 2023 – Jun. 2024

- Reproduced and compared GAN, diffusion, and flow-based models; designed conditional guidance strategies to improve signal SNR and perceptual fidelity.